

# Optimization of an RRRP-Based Latching Mechanism for CubeSat Docking Adapter

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**Abstract**—This work is carried out as part of a course project, motivated by ongoing development in the IIT Bombay Student Satellite Program’s Mission Navika, which aims to demonstrate docking between two 3U CubeSats. The docking adapters are designed to fit within the standard “tuna can” volume of P-POD deployers, imposing strict geometric and space constraints. Within this context, this paper presents the design and optimization of a planar RRRP-based latching mechanism for docking applications. The objective is to improve tolerance to axial and lateral misalignment while ensuring physically feasible inward and downward engagement motion.

A trajectory-based optimization framework is developed that incorporates geometric constraints, ensures smooth and consistent inward motion, and maintains effective force transmission throughout the operation. The problem is formulated as a constrained nonlinear optimization problem and solved using Sequential Least Squares Programming (SLSQP). The results demonstrate that the mechanism can achieve the desired inward and downward motion while improving tolerance to axial and lateral misalignments, making it suitable for compact docking systems.

## I. INTRODUCTION

This work is carried out in the context of Mission Navika, an ongoing effort to demonstrate autonomous docking between two 3U CubeSats. Each spacecraft is equipped with a docking adapter designed to fit within the standard “tuna can” volume of a P-POD deployer, imposing strict geometric and packaging constraints on the system.

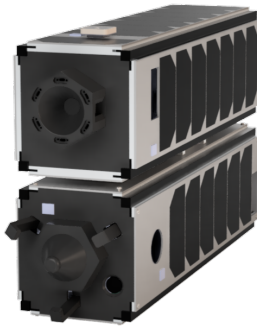


Fig. 1. CubeSats in Mission Navika

Docking between small satellites requires the mechanism to successfully engage even in the presence of positional errors between the two bodies. These errors arise due to limitations in relative navigation and control. As a result, the docking

mechanism must tolerate both axial and lateral misalignment while still achieving firm mechanical engagement.

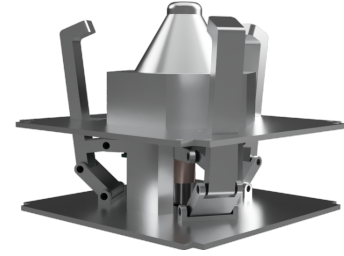


Fig. 2. Chaser DAU

As shown in fig 2 The Docking Adapter Unit (DAU) employs a three-latch architecture, where each latch is implemented as a planar RRRP linkage driven by a linear actuator. The motion of the slider produces a coupled radial and axial motion of the latch tip. The radial motion allows the latch to enter the slot of the mating interface, while the axial motion pulls the two docking ports together and generates preload.

The geometry of this linkage directly determines the trajectory of the latch tip and the force transmission characteristics of the mechanism. While a feasible linkage configuration has been obtained through preliminary design, its performance has not been systematically optimized.

## II. DOCKING MECHANISM DESCRIPTION

The docking interface geometry is assumed to be fixed in this work. This includes the latch seat location, slot geometry, mounting radius of the mechanism, and actuator stroke limits. These parameters define the allowable workspace within which the latch tip must operate.

Figure 3 shows the geometric configuration and key dimensions of the docking mechanism.

Each latch consists of a planar RRRP linkage actuated by a slider. The linkage converts the linear input motion into a nonlinear trajectory of the latch tip. The shape of this trajectory determines whether the latch can successfully enter the slot, clear surrounding features, and reach the final seating position.

The latch tip must initially possess sufficient radial clearance to enter the slot region without interference. During engagement, the latch tip must then follow a continuous inward and downward trajectory to pull the docking interfaces together. This motion must be achieved within the geometric



### A. Forward Kinematics

The slider position is parameterized by the actuator coordinate  $a_3$ , which varies between prescribed stroke limits:

$$a_{3,min} \leq a_3 \leq a_{3,max}$$

The fixed pivot point and slider point are defined as:

$$O = (R, 0)$$

$$B = (s, a_3)$$

The intermediate revolute joint  $A$  is obtained from the intersection of two circles:

- centered at  $O$  with radius  $a_1$
- centered at  $B$  with radius  $a_2$

The linkage feasibility condition therefore becomes:

$$|a_1 - a_2| \leq \|B - O\| \leq a_1 + a_2$$

The latch tip is attached to the coupler link using local coordinates  $(u, v)$ . The tip position is computed as:

$$T = A + u\hat{t} + v\hat{n}$$

where  $\hat{t}$  is the unit vector along link  $AB$  and  $\hat{n}$  is the corresponding perpendicular unit vector.

The latch tip trajectory is generated by evaluating the mechanism over the complete actuator stroke.

### B. Performance Objectives

The optimization objective is formulated as a weighted combination of multiple trajectory-based performance metrics.

1) *Closed Position Accuracy*: At the closed configuration, the latch tip should reach the desired seating position:

$$(R_{seat}, H_{seat})$$

The corresponding objective term is:

$$J_{closed} = (x_c - R_{seat})^2 + (y_c - H_{seat})^2$$

where  $(x_c, y_c)$  denotes the latch tip position in the closed configuration.

2) *Capture Capability*: The open configuration must provide sufficient radial clearance for entry into the docking slot.

$$J_{capture} = -(x_{open} - R_{hex})$$

Maximizing this quantity improves tolerance to lateral misalignment during docking.

3) *Axial Tolerance*: The trajectory should maintain sufficient vertical clearance while crossing the slot entry boundary at  $x = R_{hex}$ .

$$J_{axial} = -(y_{entry} - H_{seat})$$

where  $y_{entry}$  denotes the interpolated latch tip height at the slot entry boundary  $x = R_{hex}$ . This improves tolerance to axial offset between the docking interfaces.

4) *Monotonic Motion*: The latch trajectory should remain continuously inward and downward throughout engagement. Reversal in either direction is penalized using:

$$J_{mono} = \sum \max(0, \Delta x)^2 + \sum \max(0, \Delta y)^2$$

5) *Trajectory Smoothness*: Abrupt changes in trajectory shape are discouraged using a discrete second-derivative penalty:

$$J_{smooth} = \|\Delta^2 T\|^2$$

6) *Force Transmission Metric*: A kinematic metric related to motion transmission is incorporated to discourage configurations with poor mechanical advantage. The implemented metric favors downward motion with lower latch tip velocity relative to actuator motion.

$$v_{tip} = \sqrt{\left(\frac{dx}{da_3}\right)^2 + \left(\frac{dy}{da_3}\right)^2}$$

$$J_{force} = -\text{mean}\left(\frac{-dy/da_3}{v_{tip}^2}\right)$$

This metric favors configurations that produce larger downward latch motion for smaller actuator displacement, thereby discouraging poor mechanical advantage configurations.

### C. Constraint Formulation

Several geometric and feasibility constraints are imposed during optimization.

1) *Radial Entry Constraint*: The open configuration must satisfy:

$$x_{open} \geq R_{hex}$$

2) *Workspace Constraints*: The latch tip trajectory must remain within the allowable docking region:

$$x \geq R_{seat}$$

$$H_{seat} \leq y \leq H_{dock,t}$$

3) *Packaging Constraints*: The mechanism dimensions must remain within the available radial footprint:

$$a_1 \leq R_{max} - R$$

Additional spacing constraints are imposed to avoid interference between the linkage and surrounding hardware.

4) *Reachability Constraints*: To ensure physical assembly of the linkage throughout the stroke:

$$(a_1 + a_2) \geq d_{max}$$

$$|a_1 - a_2| \leq d_{min}$$

where  $d_{max}$  and  $d_{min}$  represent the maximum and minimum separation between points  $O$  and  $B$  over the slider stroke.

### D. Overall Objective Function

The overall optimization objective is expressed as:

$$J_{total} = 20J_{closed} + 22J_{capture} + 15J_{axial} + 15J_{mono} + 2J_{smooth} + 8J_{force} + 50P \quad (1)$$

where  $P$  represents the aggregate smooth penalty associated with geometric and feasibility constraint violations.

### E. Optimization Method

The optimization problem is solved using Sequential Least Squares Programming (SLSQP) implemented in Python using the SciPy optimization library. SLSQP is a gradient-based constrained optimization method that is well suited for nonlinear problems involving bounded design variables, nonlinear equality and inequality constraints, and weighted objective functions.

In the present work, the RRRP latch synthesis problem involves strongly coupled nonlinear geometric relationships

arising from the mechanism kinematics. The latch trajectory, workspace feasibility, and motion characteristics all vary nonlinearly with the linkage parameters. As a result, direct analytical optimization is not feasible, and numerical optimization techniques are required.

The optimizer iteratively updates the design vector

$$\mathbf{x} = (a_1, a_2, R, s, u, v)$$

to minimize the total objective function while satisfying feasibility constraints. At every iteration:

- the complete latch trajectory is generated using forward kinematics,
- geometric feasibility of the mechanism is checked,
- trajectory-based performance metrics are evaluated,
- and the total objective value is computed.

Most geometric and behavioral requirements are incorporated using smooth quadratic penalty functions rather than discontinuous logical constraints. For example, a constraint of the form

$$x \geq x_{min}$$

is converted into the penalty term

$$P(x) = \max(0, x_{min} - x)^2$$

which evaluates to zero when the constraint is satisfied and increases smoothly when violated.

This formulation provides two important advantages. First, the objective function remains continuous and differentiable, which is important for gradient-based optimization methods such as SLSQP. Second, smooth penalties avoid abrupt changes in objective value that can produce flat regions or unstable convergence behavior during optimization.

The overall optimization objective is therefore expressed as a weighted sum of trajectory performance metrics and smooth penalty terms:

$$J_{total} = \sum_i w_i J_i + \sum_j \lambda_j P_j$$

where  $J_i$  denotes the performance objectives,  $P_j$  denotes penalty functions associated with geometric violations, and  $w_i$  and  $\lambda_j$  are weighting coefficients.

The weighting coefficients used in the objective function were selected empirically based on the relative functional importance of each performance metric and iterative observation of optimizer behavior. Higher weights were assigned to quantities directly related to successful docking, such as seating accuracy, capture capability, and monotonic engagement motion. Lower weights were assigned to secondary trajectory refinement terms such as smoothness.

Penalty weights associated with geometric feasibility were intentionally chosen to be significantly larger than the performance objective weights in order to strongly discourage physically infeasible configurations during optimization.

Several trial optimizations were performed to obtain a balance between convergence stability and physically meaningful mechanism behavior. The selected weights were found to consistently produce feasible trajectories with stable inward and downward engagement motion.

In addition to penalty-based constraints, explicit nonlinear feasibility constraints are imposed to ensure that the linkage remains physically assemblable throughout the slider stroke.

These constraints are derived from the triangle inequality conditions of the RRRP linkage geometry.

The optimization process therefore combines nonlinear kinematic modeling, trajectory evaluation, smooth penalty methods, and constrained gradient-based optimization within a unified numerical framework.

## V. RESULTS

The optimization framework successfully converged to a feasible RRRP linkage configuration satisfying the geometric, kinematic, and trajectory-based requirements imposed by the docking interface.

The optimized mechanism parameters obtained from the SLSQP optimization are listed in Table I.

TABLE I  
OPTIMIZED MECHANISM PARAMETERS

| Parameter | Value (mm) |
|-----------|------------|
| $a_1$     | 14.039     |
| $a_2$     | 20.635     |
| $R$       | 25.000     |
| $s$       | 20.000     |
| $u$       | 38.372     |
| $v$       | -41.745    |

The resulting latch tip trajectory is shown in Figure 5. The trajectory satisfies the required entry clearance at the open configuration and reaches the desired seating position at the closed configuration while maintaining continuous inward and downward motion.

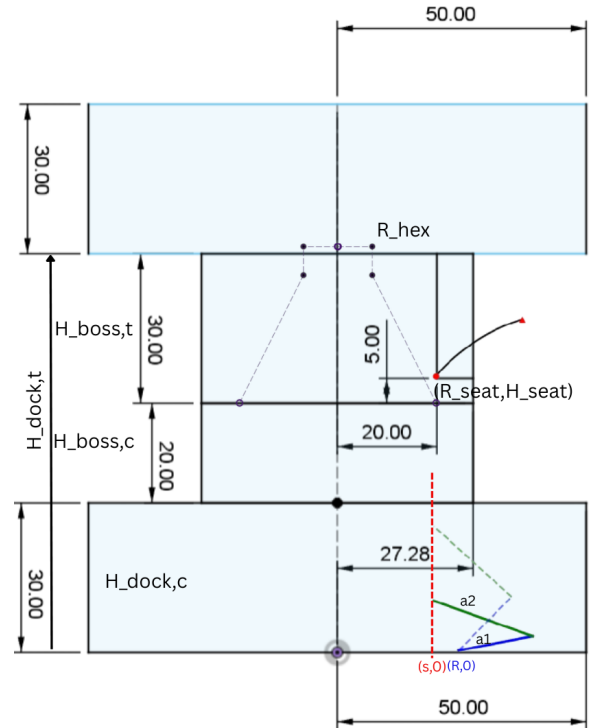


Fig. 5. Optimized Latch Tip Trajectory

Key trajectory and docking performance metrics are summarized in Table II.

TABLE II  
PERFORMANCE METRICS

| Metric                 | Value     |
|------------------------|-----------|
| Seat target error      | 0.510 mm  |
| Lateral capture margin | 9.931 mm  |
| Open tip height        | 66.964 mm |
| Axial tolerance        | 7.268 mm  |
| Maximum reach required | 25.495 mm |

The optimized trajectory demonstrates successful entry into the docking region while remaining within the prescribed geometric workspace. The latch tip trajectory crosses the slot entry boundary at a height greater than the seating height, thereby providing additional axial tolerance during engagement.

The optimization also successfully enforced monotonic inward and downward motion throughout the trajectory. No reversal or zig-zag behavior was observed in the optimized solution, indicating stable engagement behavior and reduced likelihood of jamming.

The reachability constraints remained satisfied over the entire slider stroke, ensuring that the linkage remained physically realizable throughout the motion.

## VI. DISCUSSION

The results demonstrate that trajectory-based optimization provides a systematic approach for improving docking latch performance under geometric constraints. Rather than constraining the mechanism only at discrete precision points, the present formulation evaluates the full latch trajectory throughout the actuator stroke.

The optimized linkage achieves a balance between multiple competing requirements, including entry clearance, seating accuracy, axial tolerance, packaging limitations, and motion monotonicity. The optimization process also highlights the strong coupling between linkage geometry and docking behavior. Small variations in link lengths or coupler point location significantly alter the shape of the latch trajectory.

The inclusion of monotonic motion penalties was found to be particularly important. Without these penalties, the optimizer frequently produced trajectories with reversal in radial or axial direction, resulting in geometrically feasible but physically undesirable latch motion.

Similarly, the use of smooth quadratic penalty functions improved optimizer stability by avoiding discontinuous objective behavior. This allowed the gradient-based SLSQP algorithm to converge reliably despite the highly nonlinear nature of the mechanism kinematics.

Compared to the initial baseline geometry, the optimized mechanism achieved increased axial clearance at slot entry while maintaining feasible packaging dimensions.

## VII. LIMITATIONS

The present formulation considers only planar kinematic behavior and does not include structural flexibility, friction, or dynamic contact interactions during docking.

The force transmission metric used in this work is based on a kinematic approximation rather than a complete static or dynamic force analysis. Consequently, the optimization evaluates relative transmission characteristics rather than exact actuator force requirements.

Collision detection between linkage members and surrounding hardware is also not explicitly modeled. Instead, simplified geometric constraints and directional constraints are used to avoid infeasible configurations.

Additionally, since SLSQP is a local gradient-based optimization method, convergence to the global optimum cannot be guaranteed and may depend on the initial parameter guess.

## VIII. CONCLUSION

This work presented a trajectory-based optimization framework for the design of an RRRP-based docking latch mechanism for CubeSat docking applications.

The mechanism was modeled using nonlinear kinematics, and the latch synthesis problem was formulated as a constrained nonlinear optimization problem. Multiple trajectory-based objectives, including seating accuracy, capture capability, axial tolerance, monotonic motion, and motion transmission characteristics, were incorporated within a unified optimization framework.

The optimized mechanism successfully achieved continuous inward and downward engagement motion while satisfying geometric and packaging constraints imposed by the docking adapter. The resulting trajectory demonstrated improved tolerance to axial and lateral misalignment together with physically feasible linkage motion.

The presented framework provides a systematic methodology for mechanism synthesis in compact docking systems and can be extended in future work to include contact mechanics, dynamic docking behavior, structural effects, and multi-objective optimization strategies.